Budget Analysis

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1.0 Executive Summary

Keeping government budgetary spending within approved or controlled limits while meeting variance requirements is important for maintaining a healthy financial management of public funds.

An imaginary government was chosen for this analysis taking some selected ministries into consideration.

Problem Statement:

There exist unusual levels of under-utilization across segments of the government budget

Project Objective:

This project was conducted to evaluate the effectiveness of budget control, expenditure management, and financial governance that enhance public sector accountability across sixteen (16) ministries for Fiscal Year 2025 (FY2025). The key determinants of this objective are to determine:

- Ministry with the heaviest government approved budget allocation
- The total budget allocations to capital and operational expenses
- The distribution of funding sources for the budget
- The ministry's control over budgetary spending by total budget, by ministry and expense category
- Funds availability to each ministry's budget

Project Scope:

- 5 tables were used: ***ministry_master_list, expense_type, funding source, approved_budget, actutal_expenditure***
- The ministry_master_list has 16 ministries for the government
- The expense_type has 10 capital expense (CAPEX) and 10 operational expense (OPEX) accounts used
- There are 4 funding sources: federal, provincial, grants and internally generated
- The FY2025 budget was approved on December 7 2024 for the 16 ministries for both capital and operational expenses
- The actual contains 3500 transactions from January 1 2025 to December 31 2025 for all the 16 ministries

Tools Deployed:

- Microsoft Excel
- Power Query
- Excel Charts
- Performance metrics

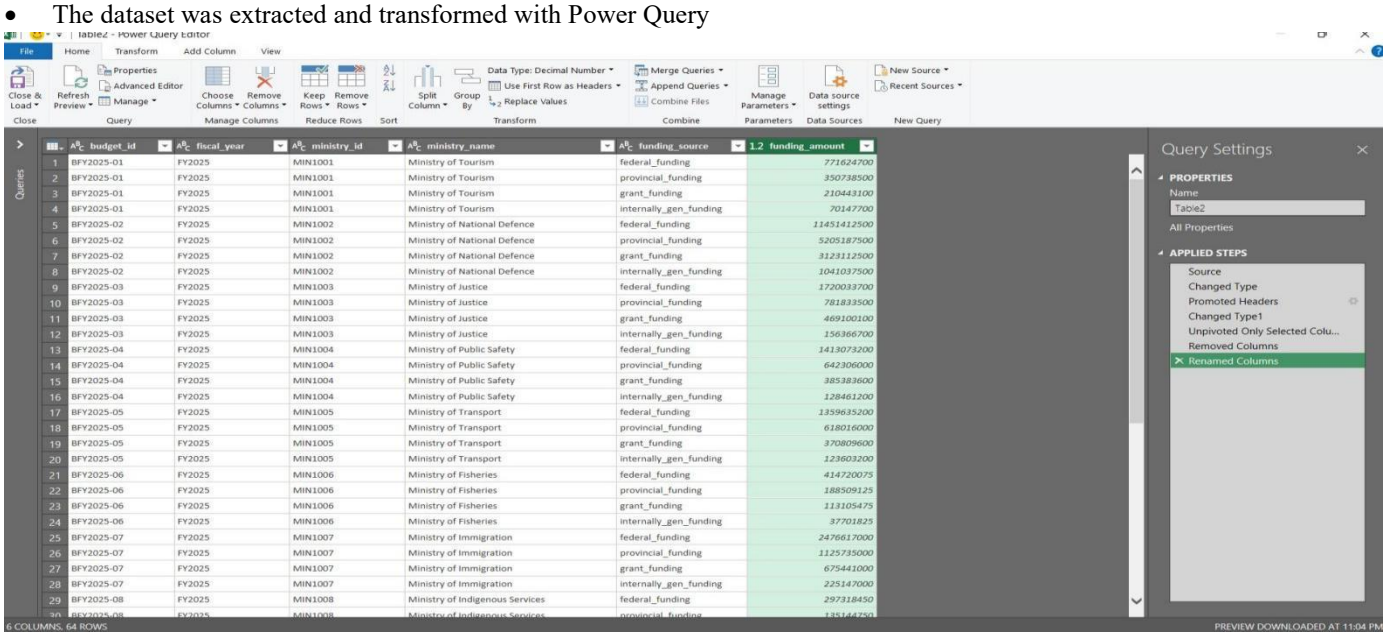
Summary of Analysis Findings

The federal government controls the FY2025 much more than the province and the entity itself through funding 55% of the budget with a major focus on defence and capital expenditure. The defence and capital allocations are 36% and 59% respectively of total allocations. The budgetary controls are strong guiding that the ministries are funded adequately and the spending behaviour falls within allocations limits. There exists common under-spending running through most of the ministries, with Ministry of Housing and Infrastructure allocated the budget. Budget management initiatives was recommended to improve the government going forward through brainstorming, regular reporting and monitoring

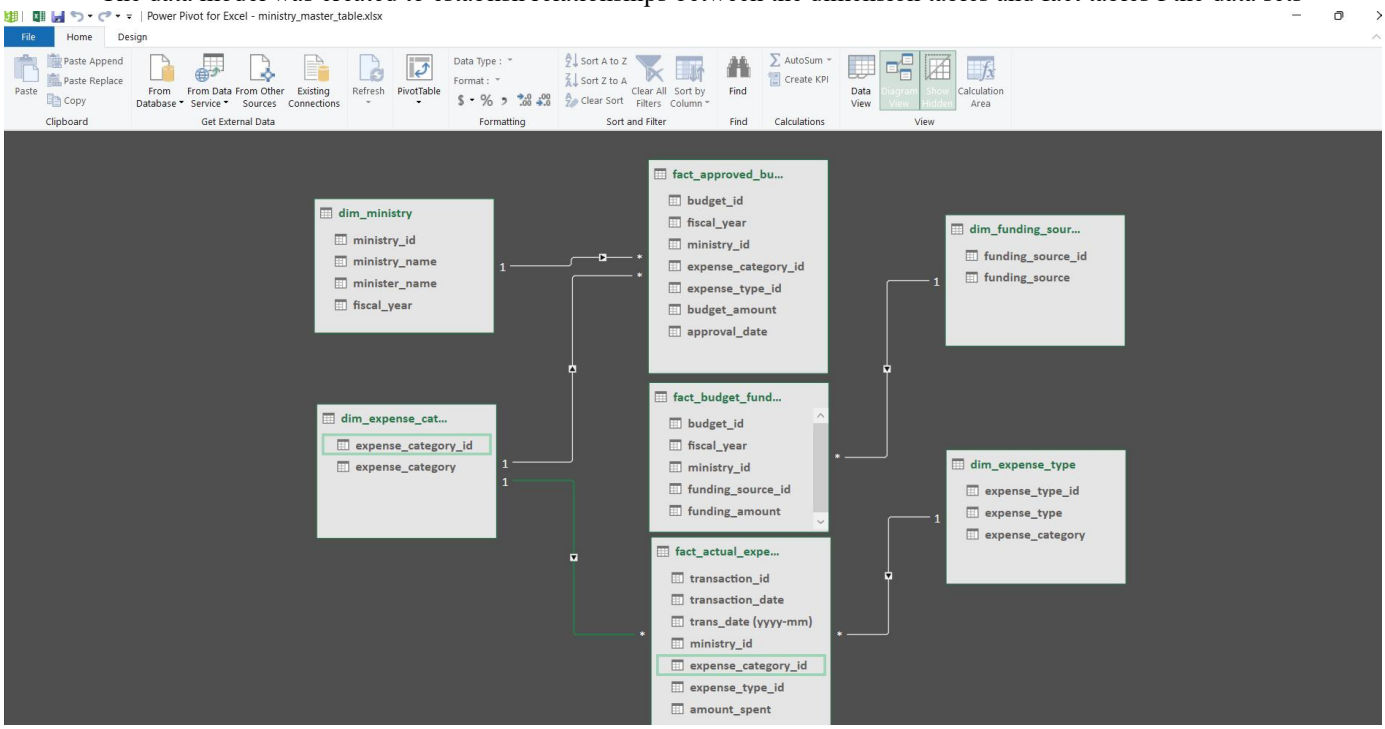
2.0 Methodology

The following methods were deployed:

- 5 tables were used: ***ministry_master_list, expense_type, funding source, approved_budget, actutal_expenditure***
  - The ministry_master table includes the columns: ***ministry_id, ministry_name, minister_name, fiscal_year***
  - The expense_type table includes columns such as: ***expense_id, fiscal_id, expense_type, expense_category***
  - Funding_source table includes: ***funding_id, funding_source, applicable_ministries***
  - The approved_budget includes: ***budget_id, fiscal_year, ministry_id, ministry_name, expense_category, expense_type, budget_amount, approval_date***
  - The actual_expenditure includes: ***transaction_id, transaction_date, ministry_id, ministry_name, expense_category, expense_type, amount_spent***
- The dataset was extracted and transformed with Power Query



- The cleaned data set was analysed using Pivot tables with descriptive interpretations
- The data model was created to establish relationships between the dimension tables and fact tables f the data sets



- Dashboard for the budget was created with Pivot Charts
- Corrective strategies were recommended to fix the identified budgetary control lapses after the analysis

Data Limitations

- **Simulated Data:** The datasets were generated by simulation and may not fully reflect real government spending or capture the spending patterns of real-world procurement cycles.
- **Controlled Overspending Scenarios:** Certain expense categories were allowed to exceed approved budgets to simulate control weaknesses.
- **Lack of Multi-Year Comparative Data:** The analysis covers FY2025 without prior year benchmarks.
- **Absence of Budget Adjustments and Supplementary Appropriations:** The data sets does not account for mid-year budget reallocations, supplementary approvals and emergency appropriation

2.0 Project Outcomes

Based on the project objective, the following key performance areas was discovered:

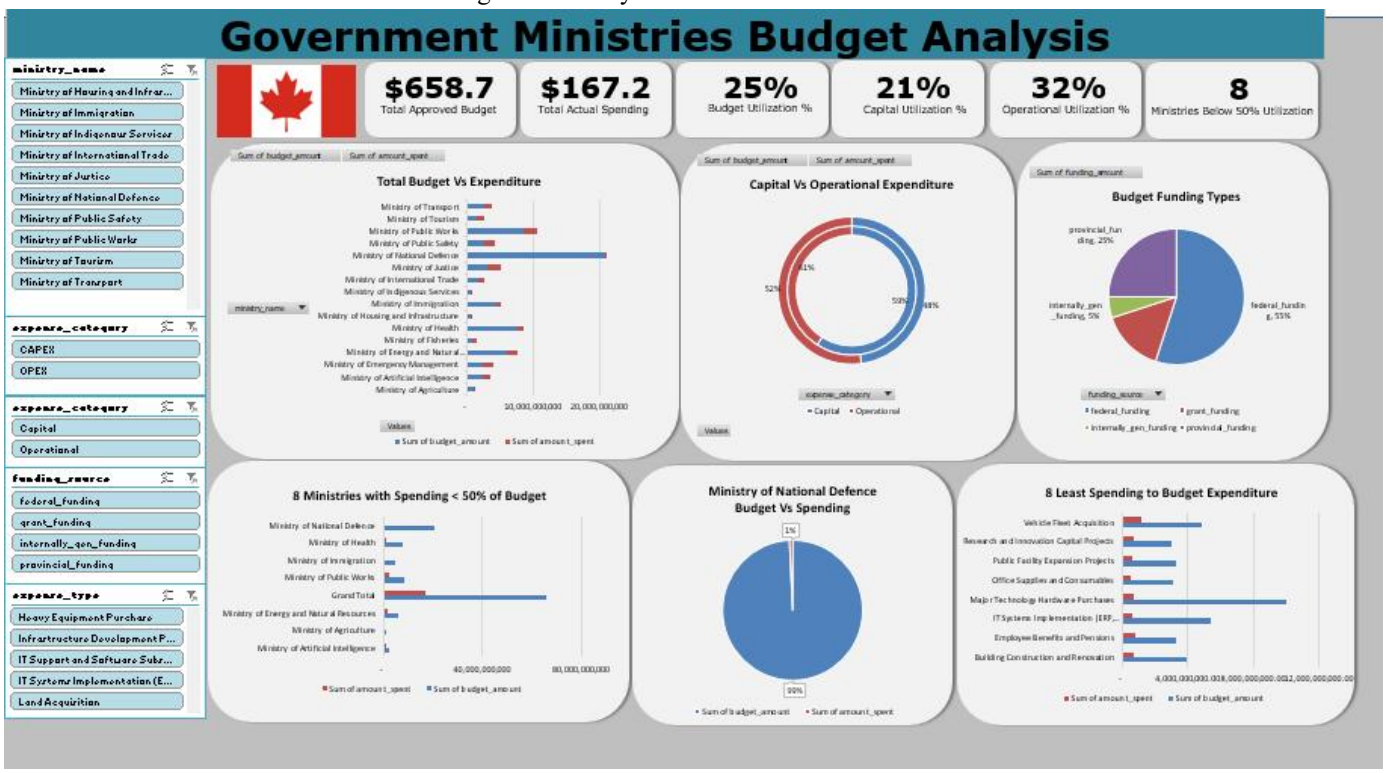
Strengths

- The Ministry of Defence has the largest budget with $20b representing 32% of the total government budget allocation
- Capital expenditure makes up 59% whiles operational expenditure makes up 41% of the total government budget allocations
- The FY2025 budget is heavily funded by the federal government with $36b representing 55% of total funding sources
- The total approved budget of $66b is adequately funded from 4 sources
- All 16 ministries were allocated funding equal to their budget allocations. No ministry was underfunded or over funded
- All 16 ministries spent within their budgetary allocations

Weaknesses

- The budget was under spent by 75% of total allocations. 25%, 21% , 32% to total spending to budget, capital spending to budget and operational spending to budget respectively. Also, 7 ministries spending below 50% of their allocations.
- The Ministry of Housing and Infrastructure has the lowest budget with $427m representing 0.6% of the total government budget allocation
- The FY2025 budget is least funded by internally generated funds of $3b representing 5% of total funding sources
- Even though the ministry of defence is the heaviest allocated ministry, it could only spend 1% of its allocation

The dashboard below summarizes the findings of the analysis.



3.0 Details of Key Objective Areas

After the government budget data was analysed in line with the stated objectives, the following information explains into details:

The budget allocations to capital and operational expense

The government allocated 59% of the total budget to capital expenditure and 41% to operational. This shows the development orientation of this budget to add to existing infrastructure across the 16 ministries. The breakdown is shown below:

Approved Budget

Row Labels	Capital	Operational	Grand Total
Grand Total	**38,897,920,000**	**26,975,055,910**	**65,872,975,910**

The distribution of funding sources for the budget

The FY2025 budget is 55% funded by the federal government and 5% by internally generated funds of each ministry. The central government is driving the direction of development with the FY2025 budget. The funding sources are shown below:

Funding Sources

Row Labels	Sum of funding_amount
federal_funding	36,230,136,750.50
provincial_funding	16,468,243,977.50
grant_funding	9,880,946,386.50
internally_gen_funding	3,293,648,795.50
Grand Total	**65,872,975,910.00**

Budgetary Control by total budget, ministries and expense category

The general budget spending pattern for FY2025 is heavily low by the government. Only 25% of the total budgetary allocation for the government was spent and 16 ministries spent within their allocated budgets even though 7 ministries could not spend beyond 50% of their budgetary allocations. The ministry of defence is the heaviest allocated ministry representing a heavy focus of government on national security but only spend 1% of its budgetary allocations. At the expense category level, 21% and 32% was spent out of allocated budget for capital and operational expenses respectively. The distribution is represented in the table below:

ministry_name	approved budget	actual spending	% spending
Ministry of Tourism	1,402,954,000.00	1,117,666,057.97	80%
Ministry of Fisheries	754,036,500.00	592,619,806.95	79%
Ministry of Emergency Management	2,343,609,210.00	1,695,694,555.12	72%
Ministry of International Trade	1,554,366,800.00	1,017,979,059.66	65%
Ministry of Justice	3,127,334,000.00	2,012,140,646.83	64%
Ministry of Housing and Infrastructure	426,926,400.00	265,459,115.73	62%
Ministry of Public Safety	2,569,224,000.00	1,555,735,018.37	61%
Ministry of Indigenous Services	540,579,000.00	297,782,131.50	55%
Ministry of Transport	2,472,064,000.00	1,276,101,377.83	52%
Ministry of Artificial Intelligence	2,409,840,000.00	1,088,331,844.68	45%
Ministry of Agriculture	871,773,000.00	343,459,253.69	39%

Ministry of Energy and Natural Resources	5,937,825,000.00	1,700,477,417.25	29%
Ministry of Public Works	8,514,730,000.00	2,039,709,319.03	24%
Ministry of Immigration	4,502,940,000.00	644,154,261.20	14%
Ministry of Health	7,624,024,000.00	918,129,366.45	12%
Ministry of National Defence	20,820,750,000.00	149,665,199.51	1%
total	**65,872,975,910.00**	**16,715,104,431.77**	**25%**

budget spending by expense category

expense_category	amount_spent	approved_budget	% spending
Capital	8,013,411,126	38,897,920,000	21%
Operational	8,701,693,306	26,975,055,910	32%
Grand Total	**16715104432**	**65872975910**	**25%**

Funds availability to budgets

The budget was 100% provided for to each ministry as represented below with the Ministry of Defence being the largest funded with almost $21b.

approved budget versus funding

Row Labels	approve_budget	funding	difference
Ministry of Agriculture	871,773,000	871,773,000	-
Ministry of Artificial Intelligence	2,409,840,000	2,409,840,000	-
Ministry of Emergency Management	2,343,609,210	2,343,609,210	-
Ministry of Energy and Natural Resources	5,937,825,000	5,937,825,000	-
Ministry of Fisheries	754,036,500	754,036,500	-
Ministry of Health	7,624,024,000	7,624,024,000	-
Ministry of Housing and Infrastructure	426,926,400	426,926,400	-
Ministry of Immigration	4,502,940,000	4,502,940,000	-
Ministry of Indigenous Services	540,579,000	540,579,000	-
Ministry of International Trade	1,554,366,800	1,554,366,800	-
Ministry of Justice	3,127,334,000	3,127,334,000	-
Ministry of National Defence	20,820,750,000	20,820,750,000	-
Ministry of Public Safety	2,569,224,000	2,569,224,000	-
Ministry of Public Works	8,514,730,000	8,514,730,000	-
Ministry of Tourism	1,402,954,000	1,402,954,000	-
Ministry of Transport	2,472,064,000	2,472,064,000	-
Grand Total	**65,872,975,910**	**65,872,975,910**	-

3.0 Strategy Recommendations

The following impacts and strategies have been recommended to aid resolve the identified government budget challenges for FY2025.

Identified Challenges	Impact	Recommendation
Significant Budget Underutilization	• Slower economic growth • Reduced public service delivery	• Require ministries to submit quarterly expenditure report for past quarter and expenditure plans for following quarter • Introduce real-time budget utilization tracking with early warning alerts for underspending ministries
• Low Allocation to Ministry of Housing and Infrastructure	• Housing shortages • Looming public infrastructure gaps	• Conducting need-based budget assessment by allocating funds based on demographic and housing demand • Leverage Public-Private Partnerships to supplement government funding in housing and infrastructure
• Low Internally Generated Funding to Support Budget	• Ministerial policy decisions may be heavily influenced by federal government which may not reflect the objective of the ministry in serving the true need of the people	• Brainstorm to discover alternative internal funding sources by each ministry
• Ministry of Defence Spending 1% of Budgetary Allocation	• Huge national security risk • Large capital projects may stall leading to cost escalation in future fiscal years	• Require feasibility, procurement and implementation readiness before approving large budgetary allocations • Release funds in tranches tied to project progress rather than upfront full year allocation